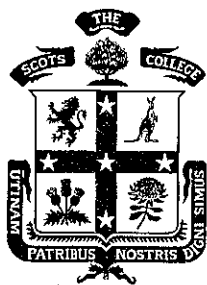


# THE SCOTS COLLEGE



**THURSDAY, 5 JUNE 2008**

## **ASSESSMENT 3**

### **YEAR 12**

# **MATHEMATICS**

### **GENERAL INSTRUCTIONS**

- Working time – 45 minutes.
- Attempt Questions 1 to 4.
- Start a new page for each Question.
- Board approved calculators may be used.
- All necessary working should be shown for every Question.

| QUESTION         | OUTCOME | MARK AVAILABLE | TOTAL     | MARK OBTAINED | TOTAL |
|------------------|---------|----------------|-----------|---------------|-------|
| 4d               | H1      | 4              | 4         |               |       |
| 3                | H4      | 5              | 5         |               |       |
| 1<br>2a<br>4a, c | H5      | 8<br>3<br>2, 3 | 16        |               |       |
| 2b<br>4b         | H9      | 3<br>2         | 5         |               |       |
|                  |         |                | <b>30</b> |               |       |

**QUESTION 1** [8 MARKS]

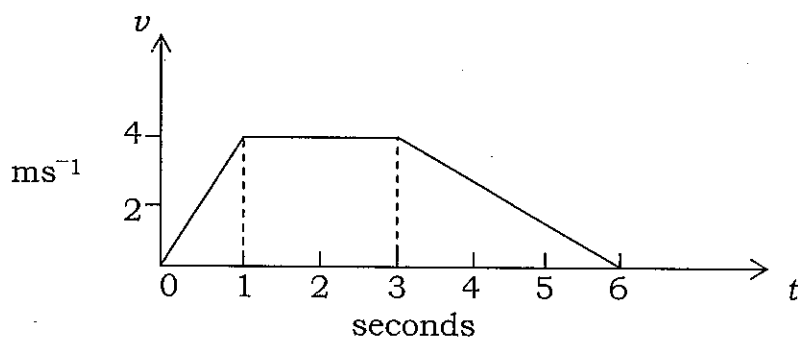
The displacement of a particle is given by the equation  $x = t^3 - 4t^2 - 3t$  where  $x$  is in metres and  $t$  is in seconds.

- (a) Find the initial velocity. [2]
- (b) Find when the particle is at rest. [2]
- (c) Find the acceleration after 4 seconds. [2]
- (d) Show that the particle is at the origin when  $t = 0$  and  $t = 2 + \sqrt{7}$ . [2]

**QUESTION 2** [6 MARKS]

- (a) The acceleration of a particle is given by  $a = -4\sin 2t$ . Initially the particle is 4m to the left of the origin and the velocity is  $2\text{cms}^{-1}$ . Find the displacement after  $\frac{\pi}{4}$  seconds. [3]

- (b) The velocity – time graph of a moving object is shown below.



- (i) Find when the object is not subject to acceleration. [1]
- (ii) Find the rate of deceleration when the object is slowing. [1]
- (iii) Find the distance the object travels in the first three seconds. [1]

**QUESTION 3** [5 MARKS]

A metal ball is cooling down according to the formula  $T = T_0 e^{-kt}$  where  $T$  is the temperature (in degrees Celsius) and  $t$  is the time in minutes. The initial temperature of the ball is  $50^\circ\text{C}$  and it cools to  $43^\circ\text{C}$  after 15 minutes.

- (a) Show that  $k = 0.01$ . [2]
- (b) Find the temperature after 1 hour. [1]
- (c) How long it takes to reach a room temperature of  $21^\circ\text{C}$  [2]

**QUESTION 4** [11 MARKS]

- (a) An arithmetic series has its sixth term equal to 18 and its eleventh term equal to 43. Find the series and write down the first three terms. [2]
- (b) Evaluate  $\sum_{n=1}^{n=8} 3^n$  [2]
- (c) An employee earns \$48,000 in their first year, with the wage increasing by 5% of the previous year's wage. Find:
  - (i) The employee's annual wage at the start of the sixth year. [1]
  - (ii) The total earnings (before tax) in the first five years. [2]
- (d) A sum of \$2,000 is invested at the start of every six months in a superannuation fund and accumulates every six months at an annual interest rate of 10%p.a.
  - (i) Find the value of the fund at the end of the first year. [1]
  - (ii) Show that at the end of  $n$  years, the fund has grown to  $\$40,000(1.05)(1.05^{2n} - 1)$ . [3]

## Standard integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE:  $\ln x = \log_e x, \quad x > 0$

# SOLUTIONS

$$x = t^3 - 4t^2 - 3t$$

$$v = \frac{dx}{dt} = 3t^2 - 8t - 3$$

$$\text{at } t=0, v = -3 \text{ m/s}$$

(b) The particle is at rest when  $v=0$

$$\therefore 3t^2 - 8t - 3 = 0$$

$$(3t+1)(t-3) = 0$$

$$\therefore t = 3, -\frac{1}{3}$$

$$\text{ie } t=3 \text{ since } t \geq 0$$

$$(c) a = \frac{dv}{dt} = 6t - 8$$

$$\text{at } t=4, a = 6(4) - 8 = 16 \text{ m/s}^2$$

$$(d) x = t^3 - 4t^2 - 3t \\ = t(t^2 - 4t - 3)$$

when  $x=0$ ,  $t=0$  and  $t^2 - 4t - 3 = 0$ .

$$\text{Solving } t^2 - 4t - 3 = 0, t = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(-3)}}{2(1)}$$

$$= \frac{4 \pm \sqrt{16+12}}{2}$$

$$= \frac{4 \pm \sqrt{28}}{2}$$

$$= \frac{4 \pm 2\sqrt{7}}{2}$$

$$= 2 \pm \sqrt{7}$$

$$\text{Since } t \geq 0, t = 2 + \sqrt{7}$$

$$\text{ie. at } x=0, t=0, 2+\sqrt{7} \text{ seconds.}$$

$$a = -4 \sin 2t$$

$$v = \int (-4 \sin 2t) dt$$

$$v = 2 \cos 2t + c_1$$

$$\text{at } t=0, v=2$$

$$\therefore 2 = 2 \cos 0 + c_1$$

$$2 = 2 + c_1 \quad \therefore c_1 = 0$$

$$v = \underline{2 \cos 2t}$$

$$x = \int 2 \cos 2t dt$$

$$x = \sin 2t + c_2$$

$$\text{at } t=0, x=-4$$

$$-4 = \sin 0 + c_2 \quad \therefore c_2 = -4$$

$$x = \sin 2t - 4$$

$$\text{at } t = \frac{\pi}{4}, x = \sin 2\left(\frac{\pi}{4}\right) - 4$$

$$= \sin \frac{\pi}{2} - 4$$

$$= 1 - 4$$

$$= -3 \text{ m}$$

(b) (i) The object is not accelerating when the velocity is constant  
This occurs between 1 and 3 seconds

(ii) The object decelerates between 3 s and 6 s. from  $4 \text{ ms}^{-1}$  to  $0 \text{ ms}^{-1}$   
 $\therefore \text{rate deceleration} = \frac{\text{speed}}{\text{time}} = \frac{4-0}{6-3}$

$$= \frac{4}{3} \text{ ms}^{-2}$$

$\text{ms}^{-2}$

$$(iii) x = \int v dt$$

$$\equiv \text{area from } 0 \text{ s to } 3 \text{ s}$$

$$= \frac{1}{2} 4(2+3)$$

$$= 10 \text{ m}$$

(area of trapezium)

$$T = T_0 e^{-kt}$$

$$\text{at } t=0, T=50.$$

$$\therefore 50 = T_0 e^{-k(0)}$$

$$50 = T_0 (1)$$

$$T_0 = 50.$$

$$(ii) \quad T = 50 e^{-kt}$$

$$43 = 50 e^{-15k}$$

$$e^{-15k} = \frac{43}{50} = 0.86$$

$$-15k = \ln 0.86$$

$$k = \frac{-\ln 0.86}{15} = 0.0100$$

For  $t=60$ ,

$$T = 50 e^{-(0.01)60}$$

$$= 27.44^\circ\text{C}$$

$$(iii) \quad T = T_0 e^{-kt}$$

$$21 = 50 e^{-(0.01)t}$$

$$e^{-0.01t} = \frac{21}{50}$$

$$\therefore -0.01t = \ln \frac{21}{50}$$

$$t = \frac{\ln 0.42}{-0.01} \text{ mins}$$

$$= 86.75 \text{ mins}$$

Q4.

$$(a) \quad T_6 = 18 = a + 5d$$

$$T_{11} = 43 = a + 10d$$

$$\therefore 5d = 25$$

$$d = 5$$

$$18 = a + 5(5)$$

$$\therefore a = -7$$

Series is  $-7, -2, 3$

$$(b) \quad \sum_{n=1}^{n=8} 3^n$$

$$= 3^1 + 3^2 + \dots + 3^8$$

$$= \frac{3(3^8 - 1)}{3 - 1}$$

$$= \frac{3}{2} (3^8 - 1)$$

$$= \frac{3}{2} (6561 - 1)$$

$$= 9840$$

$$\begin{aligned}
 (1) \quad A &= P \left(1 + \frac{r}{100}\right)^n & P &= 48,000 \\
 & & r &= 5\% \text{ pa} \\
 & & n &= 5 \text{ yrs} \\
 &= 48,000 \left(1 + \frac{5}{100}\right)^5 \\
 &= 48,000 (1.05)^5 \\
 &= \$
 \end{aligned}$$

$$\begin{aligned}
 (11) \text{ Total earnings} &= 48,000 + 48,000(1.05) + 48,000(1.05)^2 + \dots + 48,000(1.05)^5 \\
 &= 48,000 (1 + 1.05 + 1.05^2 + \dots + 1.05^5) \\
 &= 48,000 \left[ \frac{1(1.05^5 - 1)}{1.05 - 1} \right] & S_n &= a \left( \frac{r^n - 1}{r - 1} \right) \\
 &= \frac{48,000 (1.05^5 - 1)}{0.05} \\
 &= \$265,230
 \end{aligned}$$

$$(c) \quad A_n = P \left(1 + \frac{r}{100}\right)^n$$

Let  $A_n$  be the amount accumulating after  $n$  years.

$$A_{40} = 2000(1.12)^{40}$$

$$A_{39} = 2000(1.12)^{39}$$

$$A_2 = 2000(1.12)^2$$

$$A_1 = 2000(1.12)^1$$

$$\begin{aligned}
 \wedge \text{ Total amount} &= 2000 [1.12 + 1.12^2 + \dots + 1.12^{40}] \\
 &= 2000 \left[ \frac{1.12(1.12^{40} - 1)}{1.12 - 1} \right] \\
 &= \frac{2000(1.12)(1.12^{40} - 1)}{0.12} \\
 &= \$1,718,284.78
 \end{aligned}$$

Let  $A_n$  be amount of  $n$  yrs

$$(i) A_1 = 2000(1.05)$$

$$\begin{aligned}
 A_2 &= 2000(1.05) + 2000(1.05) \\
 &= 2000(1.05 + 1.05^2) \\
 &=
 \end{aligned}$$

$$(ii) A_3 = 2000(1.05 + 1.05^2 + 1.05^3)$$

Continuing this pattern

$$\begin{aligned}
 A_{2n} &= 2000(1.05 + \dots + 1.05^{2n}) \\
 &= \frac{2000(1.05)(1.05^{2n} - 1)}{0.05} \\
 &= 40,000(1.05)(1.05^{2n} - 1)
 \end{aligned}$$